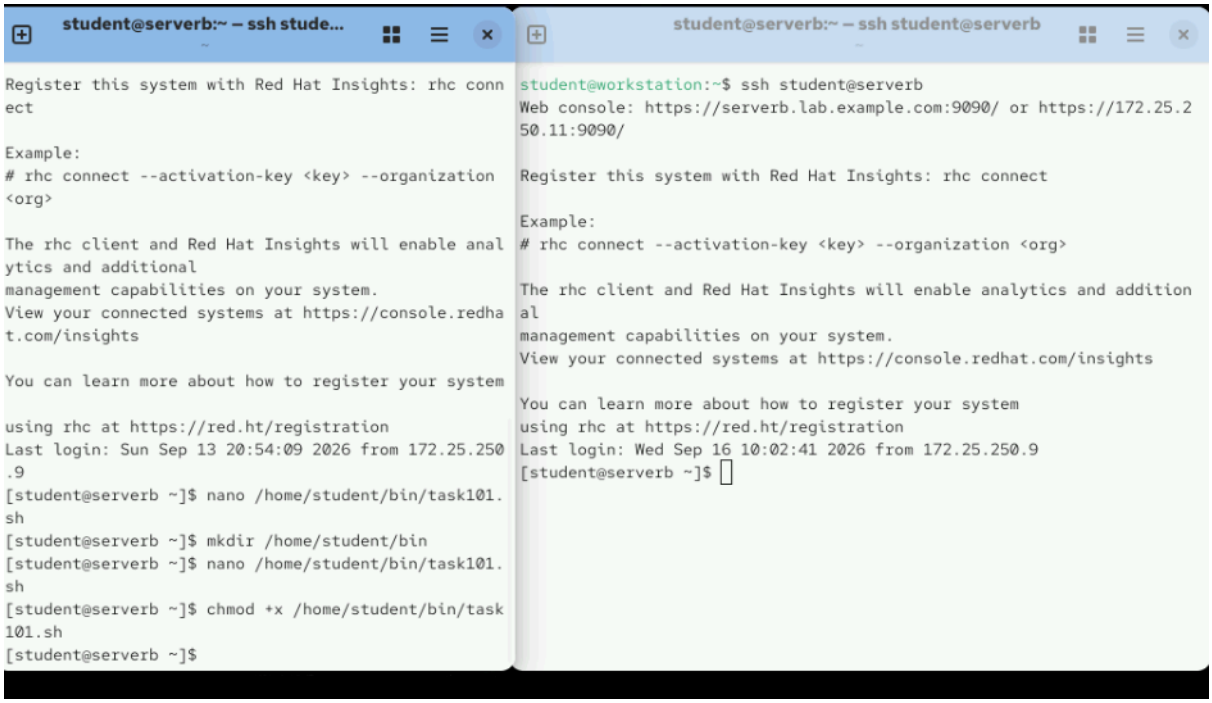
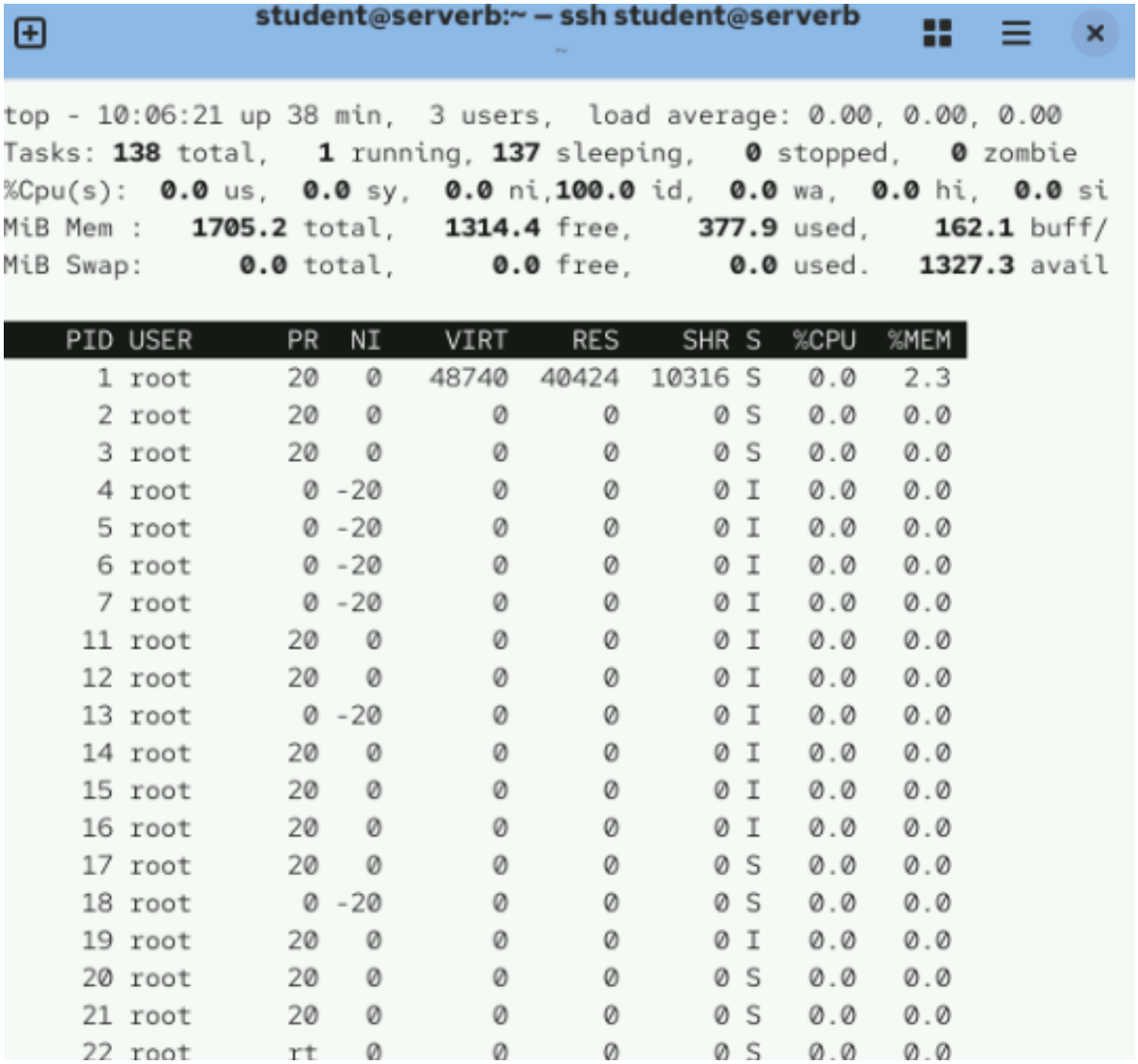
15.9-Lab: Monitor and Manage Linux Processes

Ahmed Abdelhedi

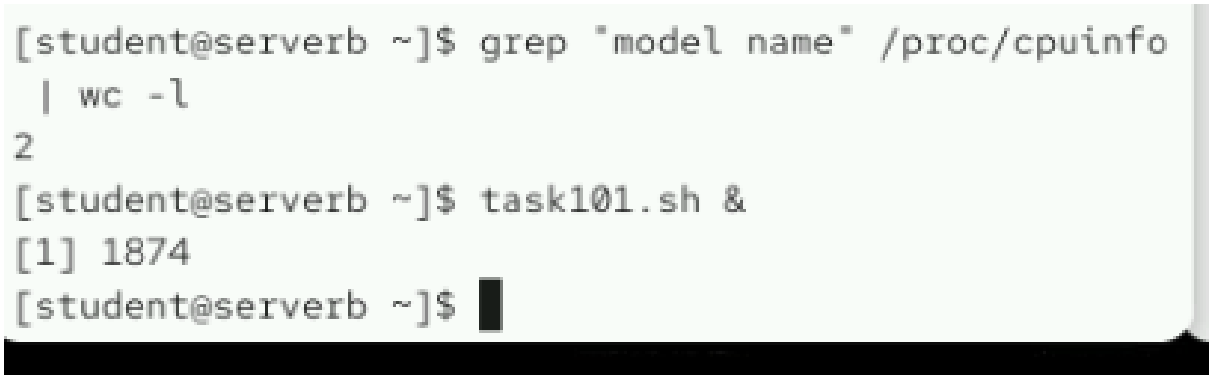
1/



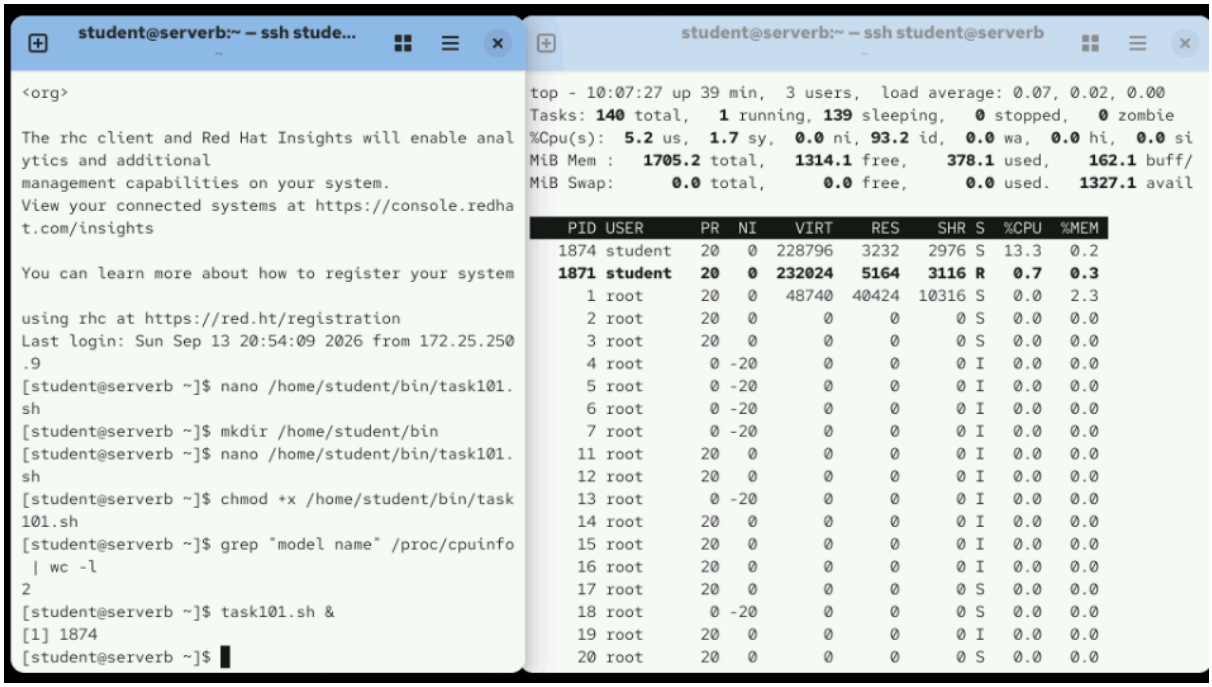
2/

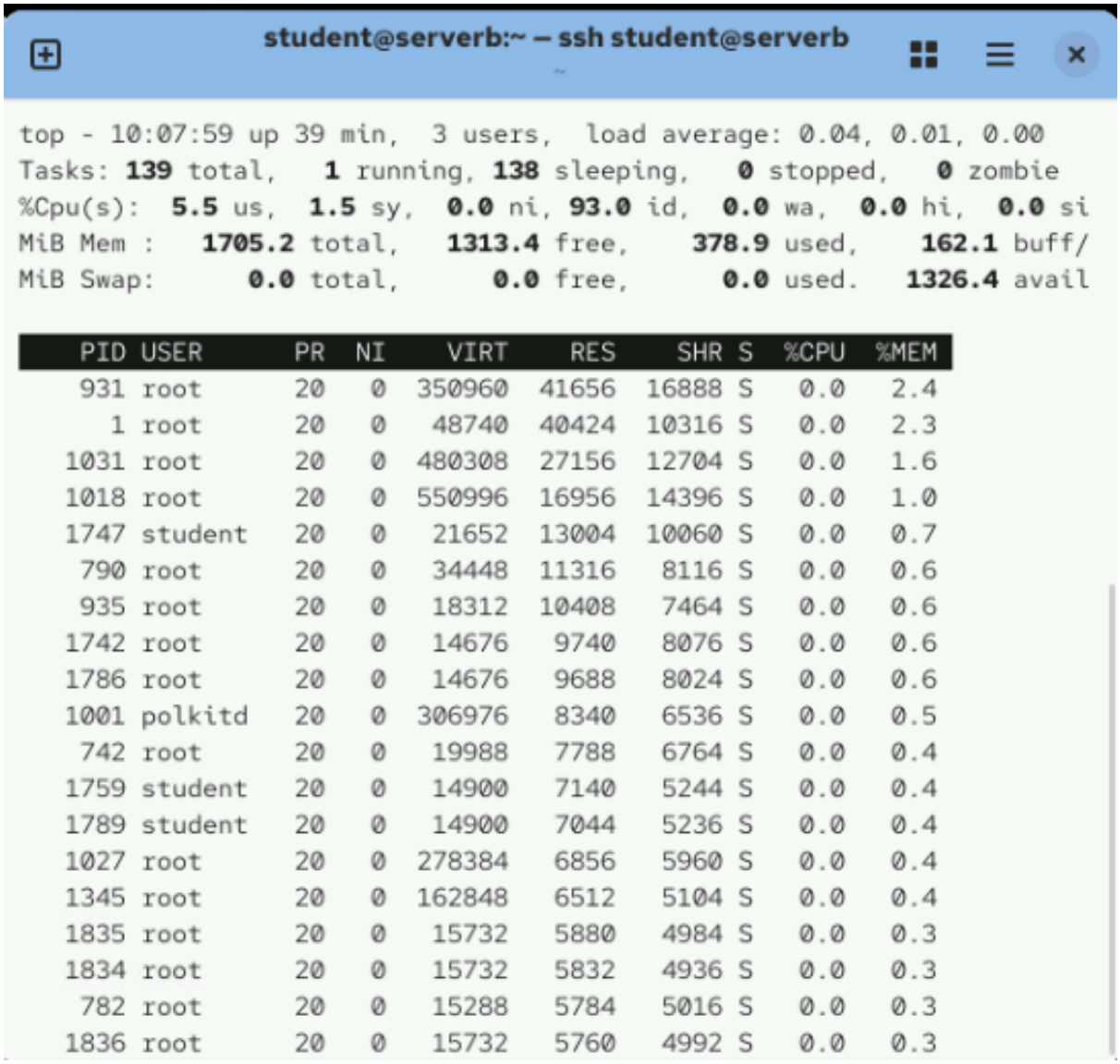

student@serverb:~ – ssh student@serverb
top - 10:06:21 up 38 min, 3 users, load average: 0.00, 0.00, 0.00
Tasks: 138 total, 1 running, 137 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 1705.2 total, 1314.4 free, 377.9 used, 162.1 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1327.3 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0
18 root 0 -20 0 0 0 S 0.0 0.0
19 root 20 0 0 0 0 I 0.0 0.0
20 root 20 0 0 0 0 S 0.0 0.0
21 root 20 0 0 0 0 S 0.0 0.0
22 root rt 0 0 0 0 S 0.0 0.0

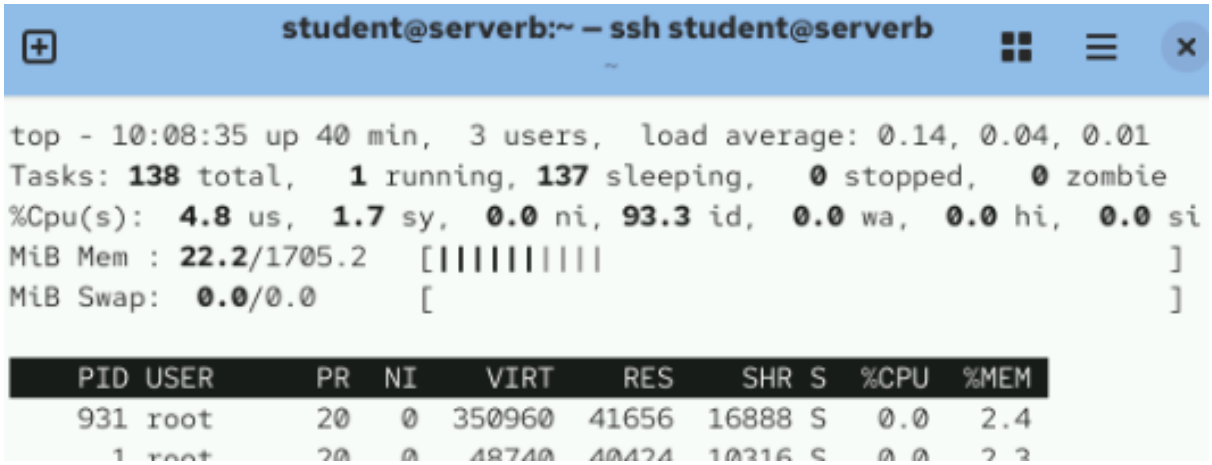
3/

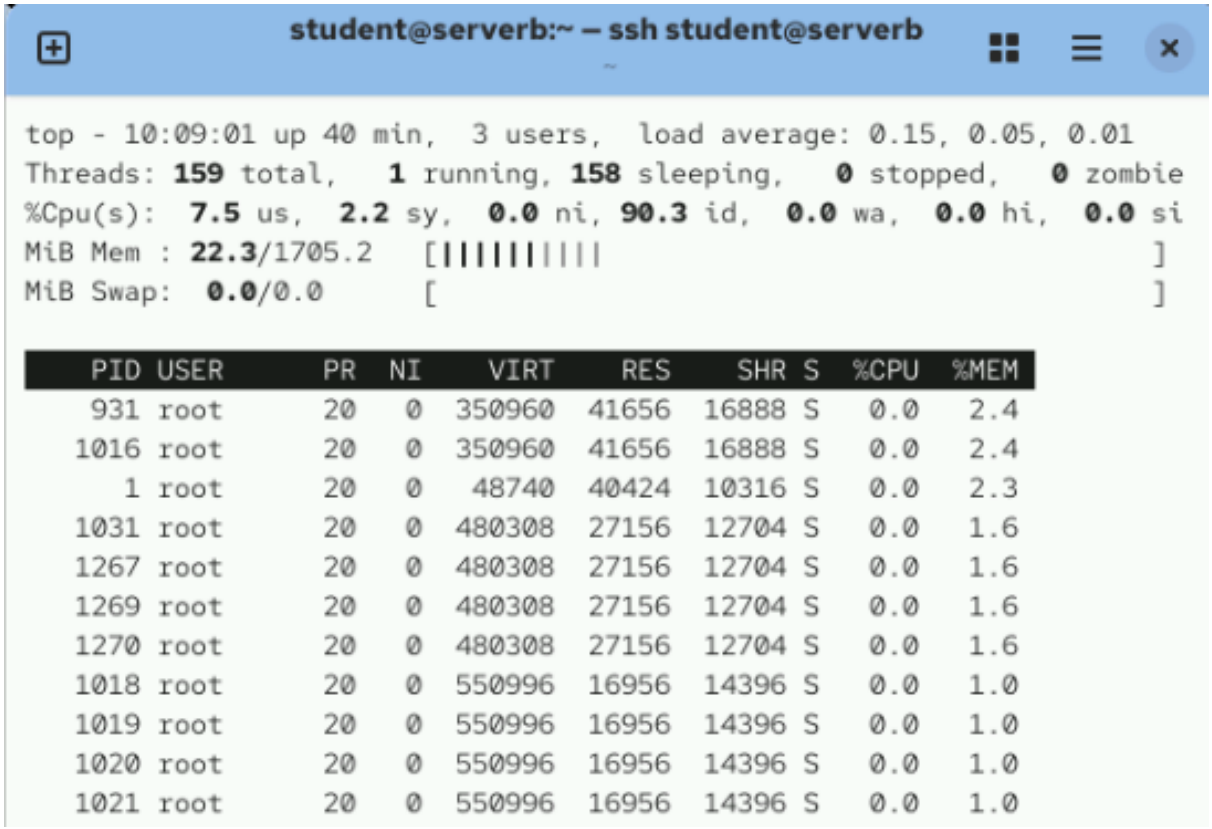

[student@serverb ~]$ grep "model name" /proc/cpuinfo
| wc -l
2
[student@serverb ~]$ task101.sh &
[1] 1874
[student@serverb ~]$

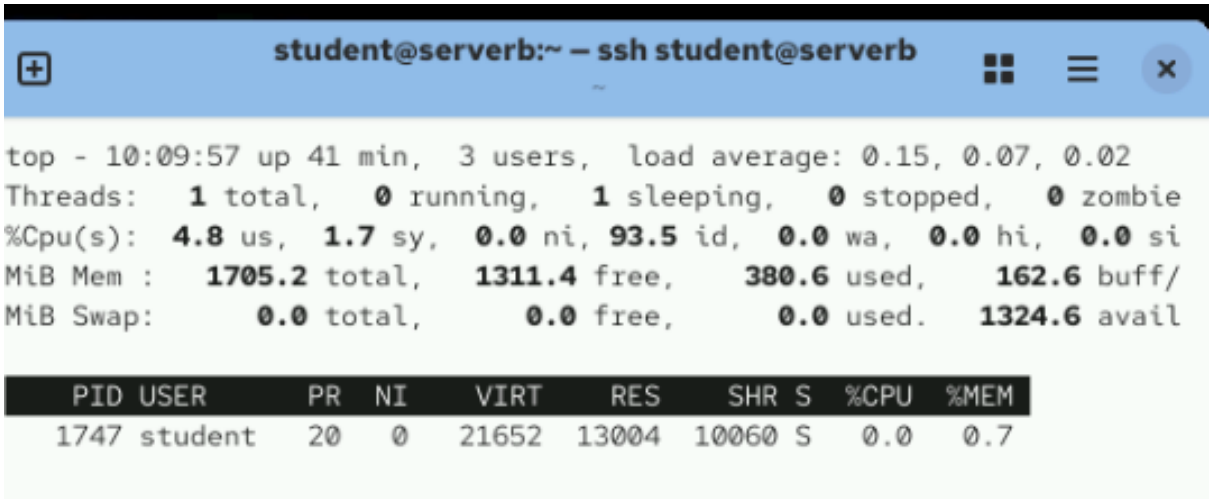
4/


student@serverb:~ — ssh stude...
<org>
The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights
You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Sun Sep 13 20:54:09 2026 from 172.25.250.9
[student@serverb ~]$ nano /home/student/bin/task101.sh
[student@serverb ~]$ mkdir /home/student/bin
[student@serverb ~]$ nano /home/student/bin/task101.sh
[student@serverb ~]$ chmod +x /home/student/bin/task101.sh
[student@serverb ~]$ grep "model name" /proc/cpuinfo | wc -l
2
[student@serverb ~]$ task101.sh &
[1] 1874
[student@serverb ~]$
student@serverb:~ — ssh student@serverb
top - 10:07:27 up 39 min, 3 users, load average: 0.07, 0.02, 0.00
Tasks: 140 total, 1 running, 139 sleeping, 0 stopped, 0 zombie
%Cpu(s): 5.2 us, 1.7 sy, 0.0 ni, 93.2 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 1705.2 total, 1314.1 free, 378.1 used, 162.1 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1327.1 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
1874 student 20 0 228796 3232 2976 S 13.3 0.2
1871 student 20 0 232024 5164 3116 R 0.7 0.3
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0
18 root 0 -20 0 0 0 S 0.0 0.0
19 root 20 0 0 0 0 I 0.0 0.0
20 root 20 0 0 0 0 S 0.0 0.0


student@serverb:~ — ssh student@serverb
top - 10:07:59 up 39 min, 3 users, load average: 0.04, 0.01, 0.00
Tasks: 139 total, 1 running, 138 sleeping, 0 stopped, 0 zombie
%Cpu(s): 5.5 us, 1.5 sy, 0.0 ni, 93.0 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 1705.2 total, 1313.4 free, 378.9 used, 162.1 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1326.4 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
931 root 20 0 350960 41656 16888 S 0.0 2.4
1 root 20 0 48740 40424 10316 S 0.0 2.3
1031 root 20 0 480308 27156 12704 S 0.0 1.6
1018 root 20 0 550996 16956 14396 S 0.0 1.0
1747 student 20 0 21652 13004 10060 S 0.0 0.7
790 root 20 0 34448 11316 8116 S 0.0 0.6
935 root 20 0 18312 10408 7464 S 0.0 0.6
1742 root 20 0 14676 9740 8076 S 0.0 0.6
1786 root 20 0 14676 9688 8024 S 0.0 0.6
1001 polkitd 20 0 306976 8340 6536 S 0.0 0.5
742 root 20 0 19988 7788 6764 S 0.0 0.4
1759 student 20 0 14900 7140 5244 S 0.0 0.4
1789 student 20 0 14900 7044 5236 S 0.0 0.4
1027 root 20 0 278384 6856 5960 S 0.0 0.4
1345 root 20 0 162848 6512 5104 S 0.0 0.4
1835 root 20 0 15732 5880 4984 S 0.0 0.3
1834 root 20 0 15732 5832 4936 S 0.0 0.3
782 root 20 0 15288 5784 5016 S 0.0 0.3
1836 root 20 0 15732 5760 4992 S 0.0 0.3


student@serverb:~ — ssh student@serverb
top - 10:08:35 up 40 min, 3 users, load average: 0.14, 0.04, 0.01
Tasks: 138 total, 1 running, 137 sleeping, 0 stopped, 0 zombie
%Cpu(s): 4.8 us, 1.7 sy, 0.0 ni, 93.3 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 22.2/1705.2 [||||||||||
MiB Swap: 0.0/0.0 [
PID USER PR NI VIRT RES SHR S %CPU %MEM
931 root 20 0 350960 41656 16888 S 0.0 2.4

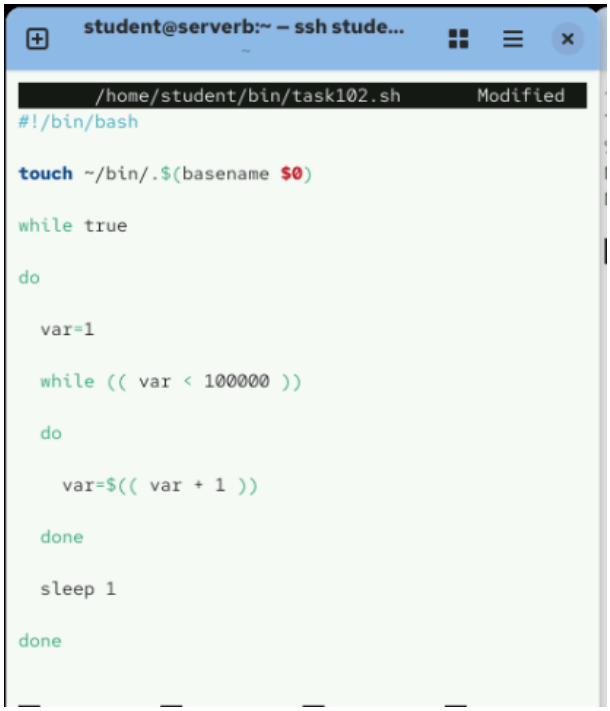

student@serverb:~ — ssh student@serverb
top - 10:09:01 up 40 min, 3 users, load average: 0.15, 0.05, 0.01
Threads: 159 total, 1 running, 158 sleeping, 0 stopped, 0 zombie
%Cpu(s): 7.5 us, 2.2 sy, 0.0 ni, 90.3 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 22.3/1705.2 [||||||||||]
MiB Swap: 0.0/0.0 [ ]
PID USER PR NI VIRT RES SHR S %CPU %MEM
931 root 20 0 350960 41656 16888 S 0.0 2.4
1016 root 20 0 350960 41656 16888 S 0.0 2.4
1 root 20 0 48740 40424 10316 S 0.0 2.3
1031 root 20 0 480308 27156 12704 S 0.0 1.6
1267 root 20 0 480308 27156 12704 S 0.0 1.6
1269 root 20 0 480308 27156 12704 S 0.0 1.6
1270 root 20 0 480308 27156 12704 S 0.0 1.6
1018 root 20 0 550996 16956 14396 S 0.0 1.0
1019 root 20 0 550996 16956 14396 S 0.0 1.0
1020 root 20 0 550996 16956 14396 S 0.0 1.0
1021 root 20 0 550996 16956 14396 S 0.0 1.0


student@serverb:~ — ssh student@serverb
top - 10:09:57 up 41 min, 3 users, load average: 0.15, 0.07, 0.02
Threads: 1 total, 0 running, 1 sleeping, 0 stopped, 0 zombie
%Cpu(s): 4.8 us, 1.7 sy, 0.0 ni, 93.5 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 1705.2 total, 1311.4 free, 380.6 used, 162.6 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1324.6 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
1747 student 20 0 21652 13004 10060 S 0.0 0.7

```
top - 10:10:16 up 42 min,  3 users,  load average: 0.17, 0.08, 0.02
Tasks: 138 total,   1 running, 137 sleeping,   0 stopped,   0 zombie
%Cpu(s):  7.6 us,  2.1 sy,  0.0 ni, 90.2 id,  0.0 wa,  0.2 hi,  0.0 si
MiB Mem :   1705.2 total,   1311.4 free,    380.6 used,    162.6 buff/
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1324.6 avail

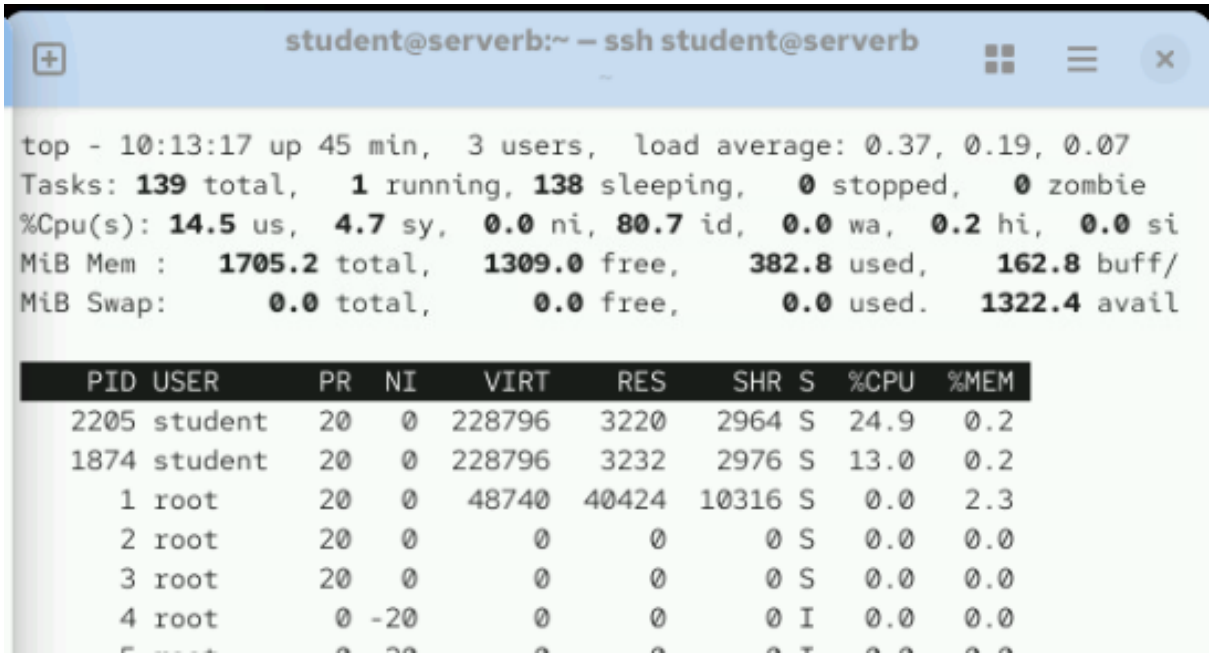
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
   1874 student   20   0  228796   3232   2976 S  19.3   0.2
   1789 student   20   0   14900   7044   5236 S   0.3   0.4
      1 root      20   0   48740  40424  10316 S   0.0   2.3
```

5/




```
[student@serverb ~]$ cp bin/task101.sh bin/task102.s
h
[student@serverb ~]$ nano /home/student/bin/task102.
sh
[student@serverb ~]$ task102.sh &
[2] 2205
[student@serverb ~]$ jobs
[1]-  Running                 task101.sh &
[2]+  Running                 task102.sh &
[student@serverb ~]$
```

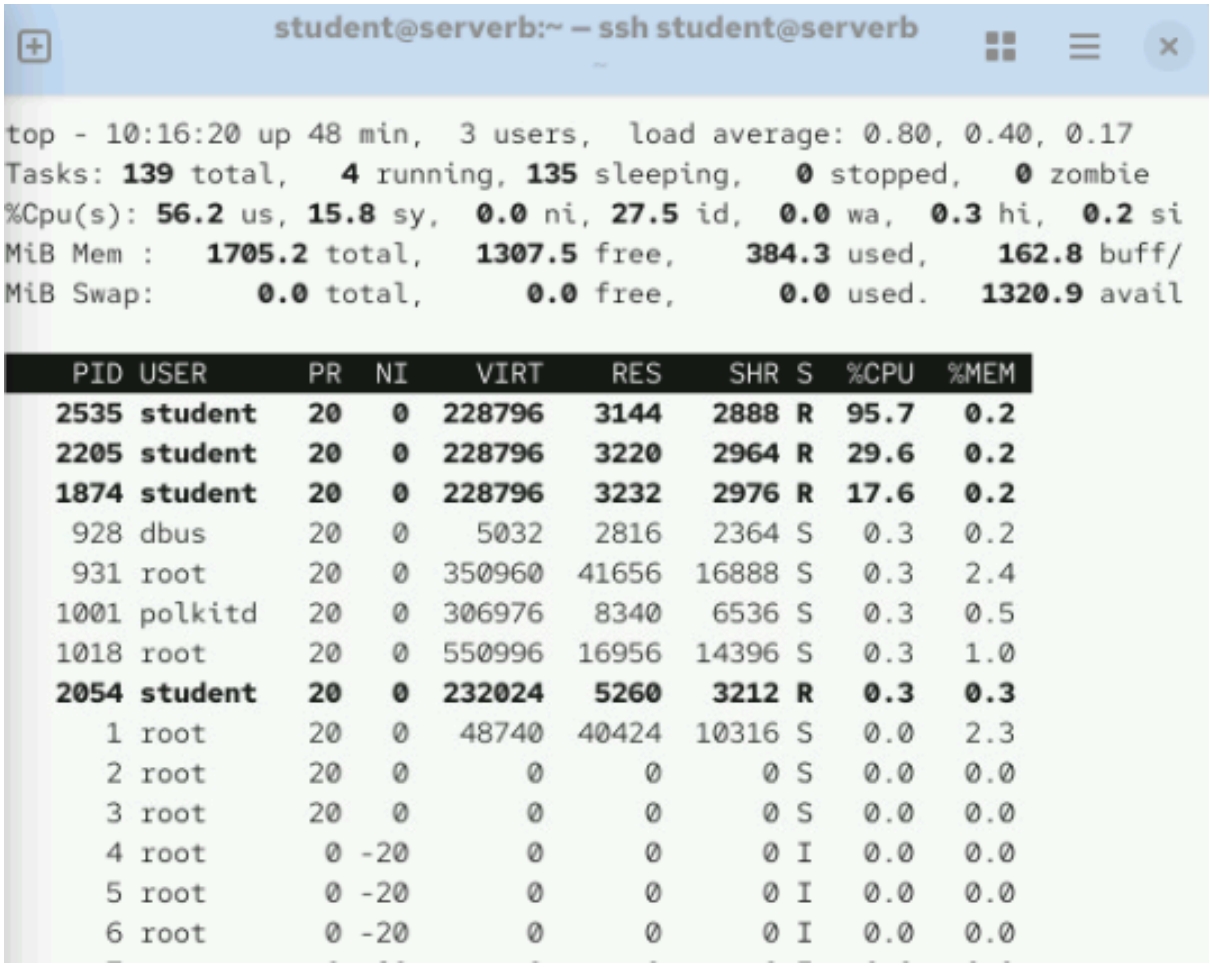
6/


student@serverb:~ – ssh student@serverb
top - 10:13:17 up 45 min, 3 users, load average: 0.37, 0.19, 0.07
Tasks: 139 total, 1 running, 138 sleeping, 0 stopped, 0 zombie
%Cpu(s): 14.5 us, 4.7 sy, 0.0 ni, 80.7 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 1705.2 total, 1309.0 free, 382.8 used, 162.8 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1322.4 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
2205 student 20 0 228796 3220 2964 S 24.9 0.2
1874 student 20 0 228796 3232 2976 S 13.0 0.2
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0

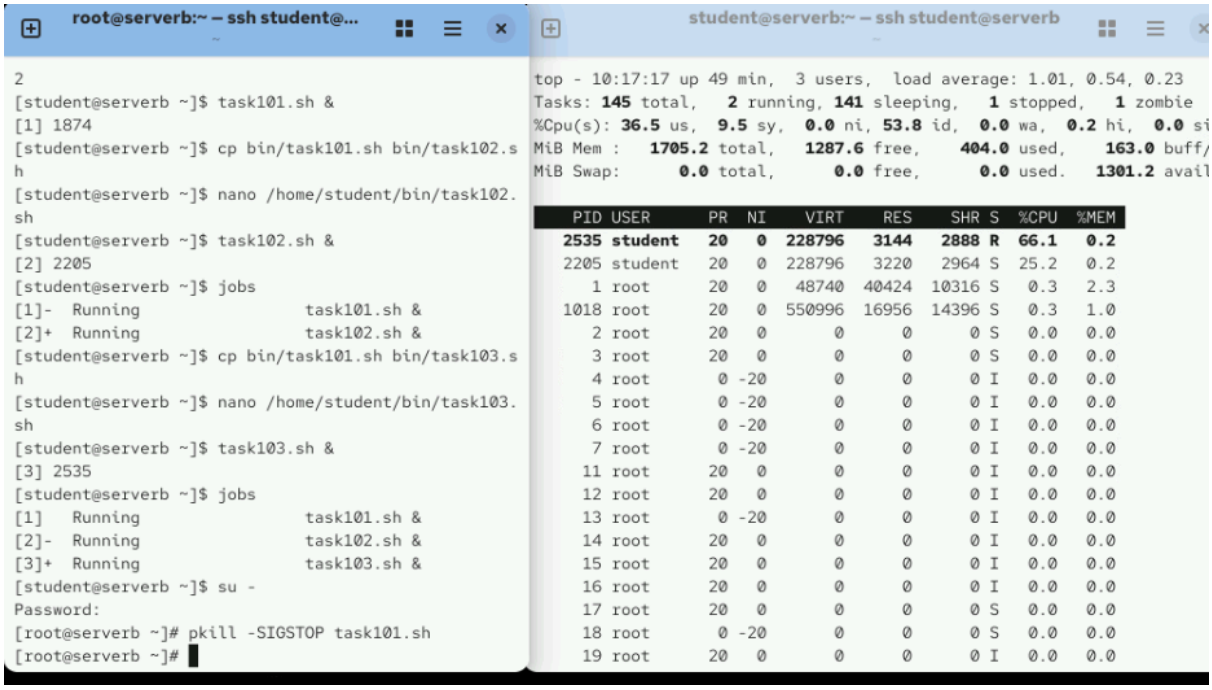
7/


student@serverb:~ – ssh stude...
/home/student/bin/task103.sh
#!/bin/bash
touch ~/bin/.$(basename $0)
while true
do
var=1
while ((var < 800000))
do
var=$((var + 1))
done
sleep 1
done
[Wrote 21 lines]
^G Help ^O Write Out ^F Where Is ^K Cut


student@serverb:~ — ssh stude...
101.sh
[student@serverb ~]$ grep "model name" /proc/cpuinfo
| wc -l
2
[student@serverb ~]$ task101.sh &
[1] 1874
[student@serverb ~]$ cp bin/task101.sh bin/task102.s
h
[student@serverb ~]$ nano /home/student/bin/task102.
sh
[student@serverb ~]$ task102.sh &
[2] 2205
[student@serverb ~]$ jobs
[1]- Running task101.sh &
[2]+ Running task102.sh &
[student@serverb ~]$ cp bin/task101.sh bin/task103.s
h
[student@serverb ~]$ nano /home/student/bin/task103.
sh
[student@serverb ~]$ task103.sh &
[3] 2535
[student@serverb ~]$ jobs
[1] Running task101.sh &
[2]- Running task102.sh &
[3]+ Running task103.sh &
[student@serverb ~]$

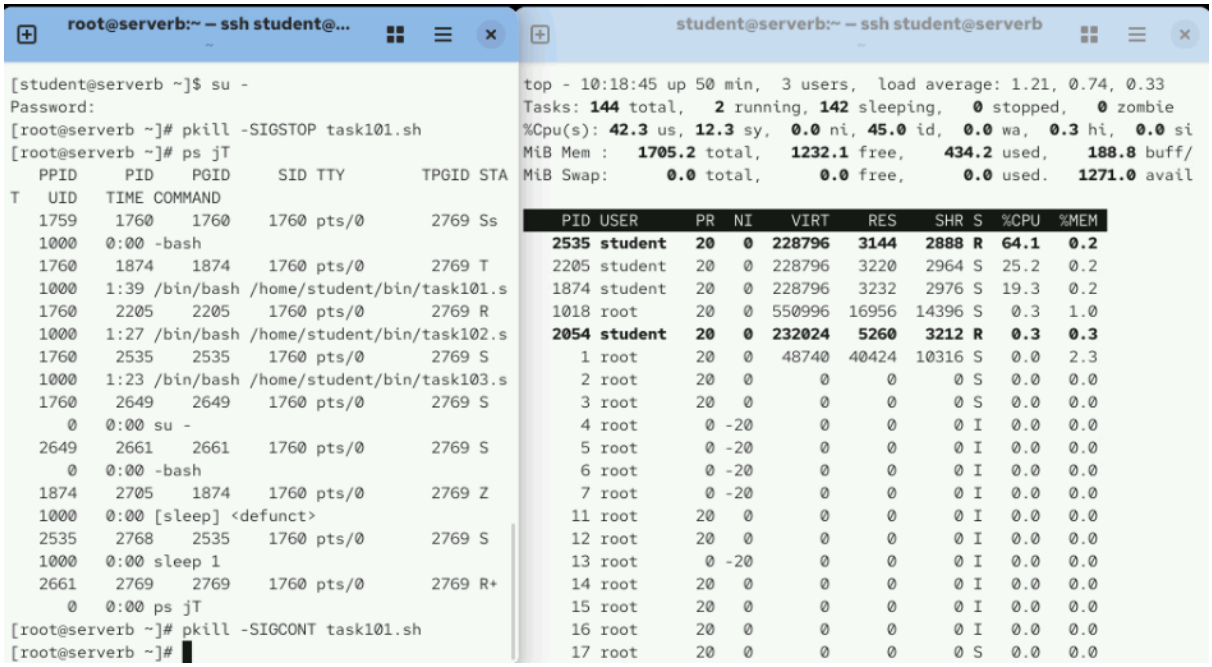

student@serverb:~ — ssh student@serverb
top - 10:16:20 up 48 min, 3 users, load average: 0.80, 0.40, 0.17
Tasks: 139 total, 4 running, 135 sleeping, 0 stopped, 0 zombie
%Cpu(s): 56.2 us, 15.8 sy, 0.0 ni, 27.5 id, 0.0 wa, 0.3 hi, 0.2 si
MiB Mem : 1705.2 total, 1307.5 free, 384.3 used, 162.8 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1320.9 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
2535 student 20 0 228796 3144 2888 R 95.7 0.2
2205 student 20 0 228796 3220 2964 R 29.6 0.2
1874 student 20 0 228796 3232 2976 R 17.6 0.2
928 dbus 20 0 5032 2816 2364 S 0.3 0.2
931 root 20 0 350960 41656 16888 S 0.3 2.4
1001 polkitd 20 0 306976 8340 6536 S 0.3 0.5
1018 root 20 0 550996 16956 14396 S 0.3 1.0
2054 student 20 0 232024 5260 3212 R 0.3 0.3
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0

8/

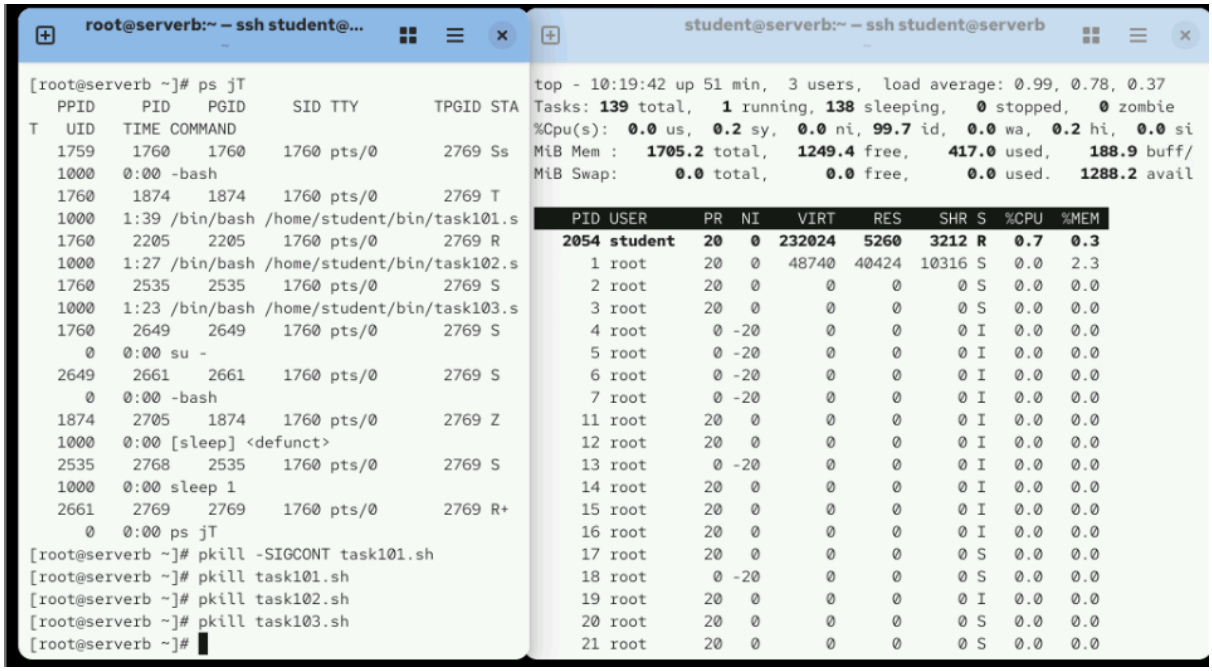



```
[root@serverb ~]# pkill -SIGSTOP task101.sh
[root@serverb ~]# ps jT
   PPID     PID    PGID     SID TTY        TPGID STA
T   UID   TIME COMMAND
   1759    1760    1760    1760 pts/0       2769 Ss
   1000   0:00 -bash
   1760    1874    1874    1760 pts/0       2769 T
   1000   1:39 /bin/bash /home/student/bin/task101.s
   1760    2205    2205    1760 pts/0       2769 R
   1000   1:27 /bin/bash /home/student/bin/task102.s
   1760    2535    2535    1760 pts/0       2769 S
   1000   1:23 /bin/bash /home/student/bin/task103.s
   1760    2649    2649    1760 pts/0       2769 S
      0   0:00 su -
   2649    2661    2661    1760 pts/0       2769 S
      0   0:00 -bash
   1874    2705    1874    1760 pts/0       2769 Z
   1000   0:00 [sleep] <defunct>
   2535    2768    2535    1760 pts/0       2769 S
   1000   0:00 sleep 1
   2661    2769    2769    1760 pts/0       2769 R+
      0   0:00 ps jT
[root@serverb ~]#
```

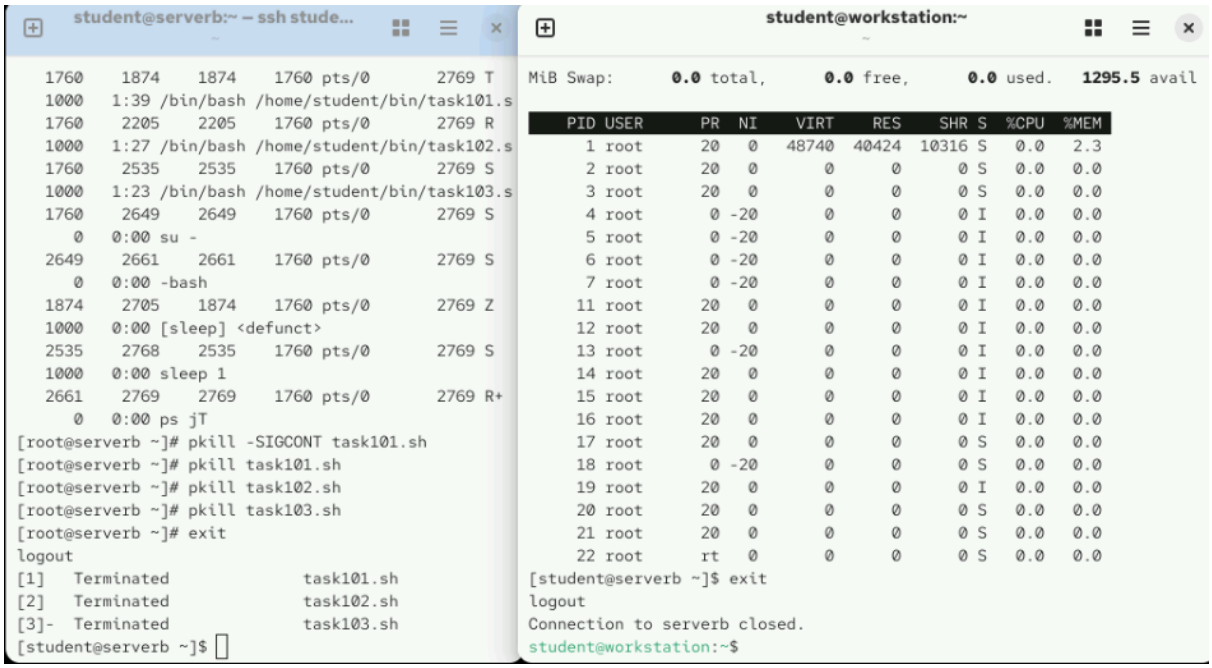
9/

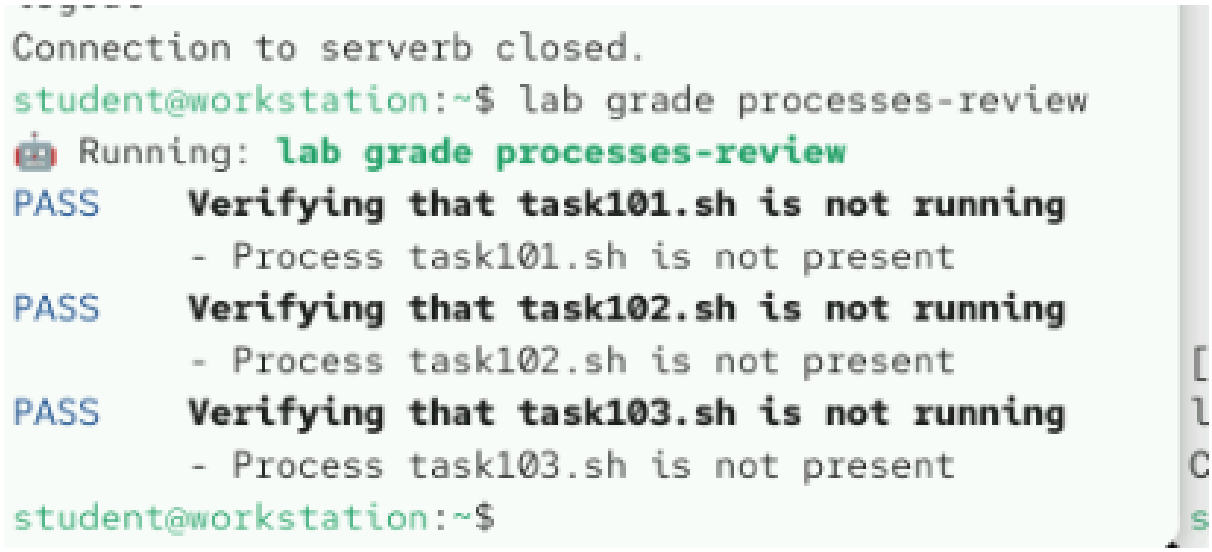

root@serverb:~ – ssh student@...
[student@serverb ~]$ su -
Password:
[root@serverb ~]# pkill -SIGSTOP task101.sh
[root@serverb ~]# ps jT
PPID PID PGID SID TTY TPGID STA
T UID TIME COMMAND
1759 1760 1760 1760 pts/0 2769 Ss
1000 0:00 -bash
1760 1874 1874 1760 pts/0 2769 T
1000 1:39 /bin/bash /home/student/bin/task101.s
1760 2205 2205 1760 pts/0 2769 R
1000 1:27 /bin/bash /home/student/bin/task102.s
1760 2535 2535 1760 pts/0 2769 S
1000 1:23 /bin/bash /home/student/bin/task103.s
1760 2649 2649 1760 pts/0 2769 S
0 0:00 su -
2649 2661 2661 1760 pts/0 2769 S
0 0:00 -bash
1874 2705 1874 1760 pts/0 2769 Z
1000 0:00 [sleep] <defunct>
2535 2768 2535 1760 pts/0 2769 S
1000 0:00 sleep 1
2661 2769 2769 1760 pts/0 2769 R+
0 0:00 ps jT
[root@serverb ~]# pkill -SIGCONT task101.sh
[root@serverb ~]#
student@serverb:~ – ssh student@serverb
top - 10:18:45 up 50 min, 3 users, load average: 1.21, 0.74, 0.33
Tasks: 144 total, 2 running, 142 sleeping, 0 stopped, 0 zombie
%Cpu(s): 42.3 us, 12.3 sy, 0.0 ni, 45.0 id, 0.0 wa, 0.3 hi, 0.0 si
MiB Mem : 1705.2 total, 1232.1 free, 434.2 used, 188.8 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1271.0 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
2535 student 20 0 228796 3144 2888 R 64.1 0.2
2205 student 20 0 228796 3220 2964 S 25.2 0.2
1874 student 20 0 228796 3232 2976 S 19.3 0.2
1018 root 20 0 550996 16956 14396 S 0.3 1.0
2054 student 20 0 232024 5260 3212 R 0.3 0.3
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0

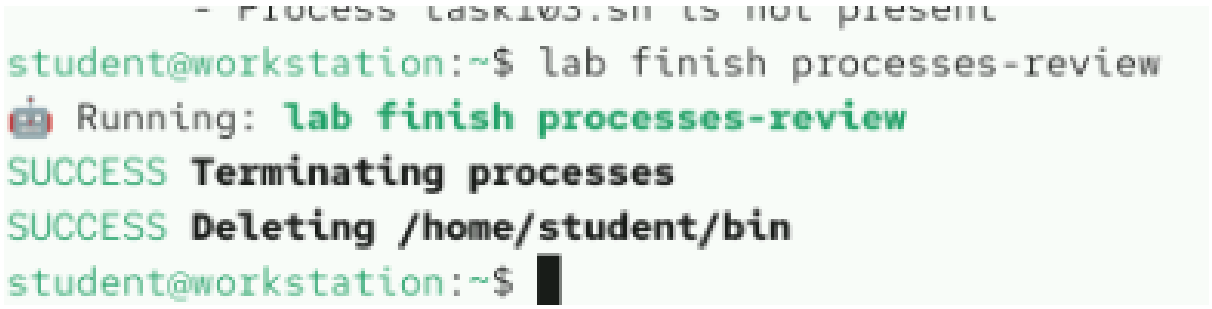
10/


root@serverb:~ – ssh student@...
[root@serverb ~]# ps jT
PPID PID PGID SID TTY TPGID STA
T UID TIME COMMAND
1759 1760 1760 1760 pts/0 2769 Ss
1000 0:00 -bash
1760 1874 1874 1760 pts/0 2769 T
1000 1:39 /bin/bash /home/student/bin/task101.s
1760 2205 2205 1760 pts/0 2769 R
1000 1:27 /bin/bash /home/student/bin/task102.s
1760 2535 2535 1760 pts/0 2769 S
1000 1:23 /bin/bash /home/student/bin/task103.s
1760 2649 2649 1760 pts/0 2769 S
0 0:00 su -
2649 2661 2661 1760 pts/0 2769 S
0 0:00 -bash
1874 2705 1874 1760 pts/0 2769 Z
1000 0:00 [sleep] <defunct>
2535 2768 2535 1760 pts/0 2769 S
1000 0:00 sleep 1
2661 2769 2769 1760 pts/0 2769 R+
0 0:00 ps jT
[root@serverb ~]# pkill -SIGCONT task101.sh
[root@serverb ~]# pkill task101.sh
[root@serverb ~]# pkill task102.sh
[root@serverb ~]# pkill task103.sh
[root@serverb ~]#
student@serverb:~ – ssh student@serverb
top - 10:19:42 up 51 min, 3 users, load average: 0.99, 0.78, 0.37
Tasks: 139 total, 1 running, 138 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.2 hi, 0.0 si
MiB Mem : 1705.2 total, 1249.4 free, 417.0 used, 188.9 buff/
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1288.2 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
2054 student 20 0 232024 5260 3212 R 0.7 0.3
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0
18 root 0 -20 0 0 0 S 0.0 0.0
19 root 20 0 0 0 0 I 0.0 0.0
20 root 20 0 0 0 0 S 0.0 0.0
21 root 20 0 0 0 0 S 0.0 0.0

11/


student@serverb:~ -- ssh stude...
1760 1874 1874 1760 pts/0 2769 T
1000 1:39 /bin/bash /home/student/bin/task101.s
1760 2205 2205 1760 pts/0 2769 R
1000 1:27 /bin/bash /home/student/bin/task102.s
1760 2535 2535 1760 pts/0 2769 S
1000 1:23 /bin/bash /home/student/bin/task103.s
1760 2649 2649 1760 pts/0 2769 S
0 0:00 su -
2649 2661 2661 1760 pts/0 2769 S
0 0:00 -bash
1874 2705 1874 1760 pts/0 2769 Z
1000 0:00 [sleep] <defunct>
2535 2768 2535 1760 pts/0 2769 S
1000 0:00 sleep 1
2661 2769 2769 1760 pts/0 2769 R+
0 0:00 ps jT
[root@serverb ~]# pkill -SIGCONT task101.sh
[root@serverb ~]# pkill task101.sh
[root@serverb ~]# pkill task102.sh
[root@serverb ~]# pkill task103.sh
[root@serverb ~]# exit
logout
[1] Terminated task101.sh
[2] Terminated task102.sh
[3]- Terminated task103.sh
[student@serverb ~]$
student@workstation:~
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1295.5 avail
PID USER PR NI VIRT RES SHR S %CPU %MEM
1 root 20 0 48740 40424 10316 S 0.0 2.3
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
11 root 20 0 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
13 root 0 -20 0 0 0 I 0.0 0.0
14 root 20 0 0 0 0 I 0.0 0.0
15 root 20 0 0 0 0 I 0.0 0.0
16 root 20 0 0 0 0 I 0.0 0.0
17 root 20 0 0 0 0 S 0.0 0.0
18 root 0 -20 0 0 0 S 0.0 0.0
19 root 20 0 0 0 0 I 0.0 0.0
20 root 20 0 0 0 0 S 0.0 0.0
21 root 20 0 0 0 0 S 0.0 0.0
22 root rt 0 0 0 0 S 0.0 0.0
[student@serverb ~]$ exit
logout
Connection to serverb closed.
student@workstation:~$


Connection to serverb closed.
student@workstation:~$ lab grade processes-review
Running: lab grade processes-review
PASS Verifying that task101.sh is not running
- Process task101.sh is not present
PASS Verifying that task102.sh is not running
- Process task102.sh is not present
PASS Verifying that task103.sh is not running
- Process task103.sh is not present
student@workstation:~$


student@workstation:~$ lab finish processes-review
Running: lab finish processes-review
SUCCESS Terminating processes
SUCCESS Deleting /home/student/bin
student@workstation:~$